WORKED SOLUTION

STEP II 2006, Question 14

Mechanics • Continuous random variable

STEP II 2006, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q14

The key idea

The whole question rests on one antiderivative. The denominator $x \ln x$ is $\ln x$ multiplied by x, and $1/x$ is the derivative of $\ln x$, so the integrand has the form $g'(x)/g(x)$ with $g(x) = \ln x$. Hence

$$\int \frac{1}{x \ln x}\,\mathrm{d}x = \ln|\ln x| + C.$$

After that, each part is a matter of applying "total probability is 1" and keeping careful track of signs and of where the density is actually non-zero.

Solution

The sketch

Write $y = \dfrac{1}{x \ln x}$ for $x > 0$, $x \neq 1$.

Sign. Since $x > 0$, y has the sign of $\ln x$: $y < 0$ on $0 < x < 1$ and $y > 0$ on $x > 1$. The curve never meets the axes.

Behaviour at the ends.

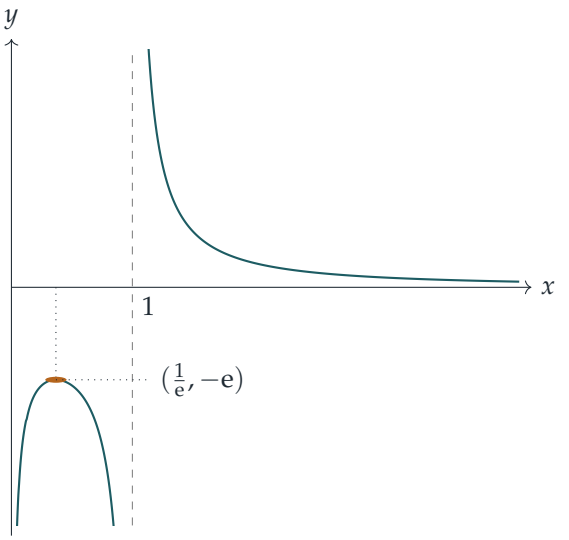
- As $x \to 0^+$: we are told $x \ln x \to 0$, and $x \ln x < 0$ there, so $y \to -\infty$.
- As $x \to 1$: $x \ln x \to 0$, negative from the left and positive from the right, so $y \to -\infty$ as $x \to 1^-$ and $y \to +\infty$ as $x \to 1^+$. The line $x = 1$ is a vertical asymptote.
- As $x \to \infty$: $x \ln x \to \infty$, so $y \to 0^+$, and the x-axis is an asymptote.

Turning points. By the quotient (or chain) rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\ln x + 1}{(x \ln x)^2},$$

which is zero only when $\ln x = -1$, that is $x = 1/\mathrm{e}$, where $y = \dfrac{1}{(1/\mathrm{e})(-1)} = -\mathrm{e}$. For

$0 < x < 1/\mathrm{e}$ we have $\ln x + 1 < 0$, so y is increasing; for $x > 1/\mathrm{e}$ (with $x \neq 1$) it is decreasing. So $(1/\mathrm{e}, -\mathrm{e})$ is a local maximum of the left-hand branch, and the right-hand branch decreases steadily from $+\infty$ to 0.



Part (i)

Here $a = \frac{1}{4}$ and $b = \frac{1}{2}$, both in $(0, 1)$, where $x \ln x < 0$. Using the antiderivative above,

$$\int_{1/4}^{1/2} \frac{\mathrm{d}x}{x \ln x} = \Big[\ln|\ln x|\Big]_{1/4}^{1/2} = \ln|\ln \tfrac{1}{2}| - \ln|\ln \tfrac{1}{4}|$$
$$= \ln(\ln 2) - \ln(2 \ln 2) = -\ln 2.$$

The total probability must be 1, so $\lambda \cdot (-\ln 2) = 1$, giving

$$\lambda = -\frac{1}{\ln 2}.$$

The negative sign is exactly what we need: $x \ln x < 0$ on $[\frac{1}{4}, \frac{1}{2}]$, so λ must be negative for f to be non-negative there.

Part (ii)

With $\lambda = 1$ the density must be non-negative, so the interval $[a, b]$ must lie where $x \ln x > 0$, which it does since $b > a > 1$. Then $\ln a, \ln b > 0$ and

$$1 = \int_a^b \frac{\mathrm{d}x}{x \ln x} = \ln(\ln b) - \ln(\ln a) = \ln \frac{\ln b}{\ln a}.$$

So $\dfrac{\ln b}{\ln a} = \mathrm{e}$, that is $\ln b = \mathrm{e} \ln a = \ln\left(a^{\mathrm{e}}\right)$, and since ln is one-to-one, $b = a^{\mathrm{e}}$, as required.

Part (iii)

With $a = \mathrm{e}$, part (ii) gives $b = \mathrm{e}^{\mathrm{e}}$. Since $1 < \frac{3}{2} < 2 < \mathrm{e}$, the whole interval $[\mathrm{e}^{3/2}, \mathrm{e}^2]$ lies inside $[a, b] = [\mathrm{e}, \mathrm{e}^{\mathrm{e}}]$, so

$$\mathrm{P}(\mathrm{e}^{3/2} \le X \le \mathrm{e}^2) = \Big[\ln(\ln x)\Big]_{\mathrm{e}^{3/2}}^{\mathrm{e}^2} = \ln 2 - \ln \tfrac{3}{2} = \ln \tfrac{4}{3}.$$

To turn this into a fraction, use the Maclaurin series $\ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \cdots$ with $t = \frac{1}{3}$, keeping four terms:

$$\ln \tfrac{4}{3} \approx \frac{1}{3} - \frac{1}{18} + \frac{1}{81} - \frac{1}{324} = \frac{108 - 18 + 4 - 1}{324} = \frac{93}{324} = \frac{31}{108}.$$

? **Ask yourself** How do we know four terms is a sensible place to stop?

The series alternates with terms decreasing in size, so the error after four terms is less than the fifth term, $\frac{1}{5 \cdot 3^5} = \frac{1}{1215} < 0.001$. So $\mathrm{P} = \ln \frac{4}{3} \approx \frac{31}{108}$, as required.

Part (iv)

Now $a = \mathrm{e}^{1/2}$, so by part (ii)

$$b = a^{\mathrm{e}} = \mathrm{e}^{\mathrm{e}/2}.$$

? **Ask yourself** Before integrating, where is the density actually non-zero?

Since $\mathrm{e} < 3$, we have $\mathrm{e}/2 < \frac{3}{2}$, so $b = \mathrm{e}^{\mathrm{e}/2} < \mathrm{e}^{3/2}$. The interval $[\mathrm{e}^{3/2}, \mathrm{e}^2]$ lies entirely to the right of b, where $\mathrm{f}(x) = 0$. Hence

$$\mathrm{P}(\mathrm{e}^{3/2} \le X \le \mathrm{e}^2) = 0.$$

(Blindly integrating $1/(x \ln x)$ from $\mathrm{e}^{3/2}$ to e^2 would give $\ln \frac{4}{3}$ again, which is wrong.)

Checks

- Numerically, $\mathrm{e}^{-1} \approx 0.368$ and a scan of y on $(0, 1)$ gives its largest value $\approx -2.718 = -\mathrm{e}$ there.
- Simpson's rule gives $\int_{1/4}^{1/2} \frac{\mathrm{d}x}{x \ln x} \approx -0.6931 = -\ln 2$, so $\lambda \approx -1.4427$.
- For several values of $a > 1$ (e.g. $a = 2$, $a = 7.3$), $\int_a^{a^{\mathrm{e}}} \frac{\mathrm{d}x}{x \ln x}$ comes out as 1.0000.
- $\ln \frac{4}{3} \approx 0.28768$ and $\frac{31}{108} \approx 0.28704$, agreeing to within 0.0007.
- In (iv), $b = \mathrm{e}^{\mathrm{e}/2} \approx 3.89$ while $\mathrm{e}^{3/2} \approx 4.48$.

Common slips

- Writing $\ln(\ln x)$ in part (i), where $\ln x < 0$: the antiderivative needs the modulus, $\ln|\ln x|$.
- Giving $\lambda = \frac{1}{\ln 2}$: a positive λ would make the "density" negative.
- In part (iv), integrating over $[e^{3/2}, e^2]$ without checking that it lies inside the support $[a, b]$.

Beyond STEP

The substitution $u = \ln x$ turns this density into $1/u$ on $[\ln a, \ln b]$, and the condition $b = a^e$ is just $\int_{\ln a}^{\ln b} \frac{du}{u} = 1$. Transforming random variables like this (here $U = \ln X$) is a central tool in first-year probability, where the density of a function of a random variable is found from the chain rule. The same function $1/(x \ln x)$ is also famous in analysis: $\sum 1/(n \ln n)$ diverges, by comparison with this integral, even though its terms tend to zero. See Grimmett and Welsh, *Probability: An Introduction*, for transformations of random variables.

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